ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSD ENGINEERING — SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain HCI foundation principles. Describe human visual and cognitive processing models. [14 Marks]

[OR]

Q.1 (Alt) Explain Norman's 7 Principles of Design: Affordance, Signifiers, Mapping, Feedback, Constraints. [14 Marks]

--- UNIT II ---

Q.2 Explain User-Centered Design (UCD) lifecycle: Empathize, Define, Ideate, Prototype, and Test. [14 Marks]

[OR]

Q.2 (Alt) Describe User Persona creation, Journey Mapping, and Information Architecture hierarchies. [14 Marks]

--- UNIT III ---

Q.3 Compare Low-Fidelity wireframing vs High-Fidelity interactive prototyping with design systems. [14 Marks]

[OR]

Q.3 (Alt) Explain Gestalt Principles of visual perception: Proximity, Similarity, Continuity, Closure. [14 Marks]

--- UNIT IV ---

Q.4 Explain Usability Evaluation methods: Nielsen's 10 Heuristics for User Interface Design. [14 Marks]

[OR]

Q.4 (Alt) Describe A/B Testing, Eye Tracking, and Think-Aloud usability testing protocols. [14 Marks]

--- UNIT V ---

Q.5 Explain WCAG 2.1 Web Accessibility Guidelines for inclusive and assistive technology design. [14 Marks]

[OR]

Q.5 (Alt) Describe Multimodal interfaces: Voice User Interfaces (VUI), Touchless gesture recognition, and AR/VR. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****